Exploratory Data Analysis with Python

A comprehensive guide to understanding, cleaning, and analyzing data through systematic exploration and visualization techniques.

Course Overview

What is Exploratory Data Analysis?

Exploratory Data Analysis (EDA) is a critical step in the data analysis process that involves summarizing and visualizing the main characteristics of a dataset. EDA helps data analysts and scientists understand the underlying patterns, anomalies, and relationships within the data before proceeding to more complex modeling techniques.

This course uses a Kabaddi match events dataset to demonstrate practical EDA techniques using Python libraries including Pandas, NumPy, Matplotlib, and Seaborn.

Why is EDA Required?

Understanding Data Distribution

Reveals central tendency, spread, and shape of data. Identifies skewness, kurtosis, and irregularities in different data types.

Identifying Outliers

Box plots and scatter plots detect anomalies that significantly impact statistical analyses and predictive models.

Checking Missing Values

Identifies patterns in missing data, essential for deciding appropriate imputation methods or handling strategies.

Uncovering Relationships

Correlation analysis and visualizations reveal relationships between variables, informing feature engineering.

Benefits of EDA

Improved Data Quality

Ensures data is clean, consistent, and accurate by detecting and correcting errors, inconsistencies, and missing values.

Better Decision Making

Provides clear understanding enabling informed decisions, identifying relevant features and potential transformations.

Informed Feature Engineering

Guides creation of new features and selection of relevant ones, enhancing predictive power of models.

Enhanced Communication

Visualizations and statistics effectively communicate findings to stakeholders, facilitating collaboration.

Critical Warning: Skipping EDA can lead to poor data quality, missed insights, and ineffective models, ultimately compromising the entire analysis process.

Case Study: Kabaddi Match Events Analysis

Objective

Perform in-depth EDA on Kabaddi match event data to gain insights into gameplay, identify patterns, and uncover relationships between match events and player performances.

Key Learning Goals

- Clean and manipulate data using Python libraries
- Create sophisticated visualizations
- Master group by operations and pivot tables
- Perform advanced aggregations and percentage calculations

Dataset Overview

The dataset contains event-level data from Kabaddi matches with comprehensive information about raids, defenses, player actions, and match outcomes.

1

Player & Team Data

Player IDs, raider IDs, defender IDs, team IDs for both raiding and defending teams

2

Event Information

Event types, timestamps (clock), event descriptions, sequential event numbers, match IDs

3

Scoring Details

Raid points, defending points, bonus points, touch points, all-out points, technical points

4

Special Events

Super raids, super tackles, do-or-die raids, substitutions, review outcomes and reasons

EDA Framework: 8 Essential Steps

01

Data Loading

Load datasets using CSV, Excel, or SQL methods. Display initial rows to understand structure.

02

Basic Exploration

Inspect data structure, types, missing values. Generate summary statistics and initial visualizations.

03

Data Cleaning

Handle missing values, inconsistencies, duplicates. Convert data types and ensure data integrity.

04

Data Transformation

Create new features, apply transformations to correct, normalize, or standardize data.

05

Outlier Handling

Identify outliers using visual and statistical methods. Decide on removal or transformation approach.

06

Descriptive Analysis

Perform univariate, bivariate, and multivariate analysis using various plots and visualizations.

07

Group By Operations

Calculate aggregated statistics based on grouping variables. Analyze data from multiple perspectives.

08

Advanced Pivot Tables

Create pivot tables with row-wise, column-wise percentages. Visualize with heatmaps to uncover patterns.

Key Technical Concepts

Data Loading Methods

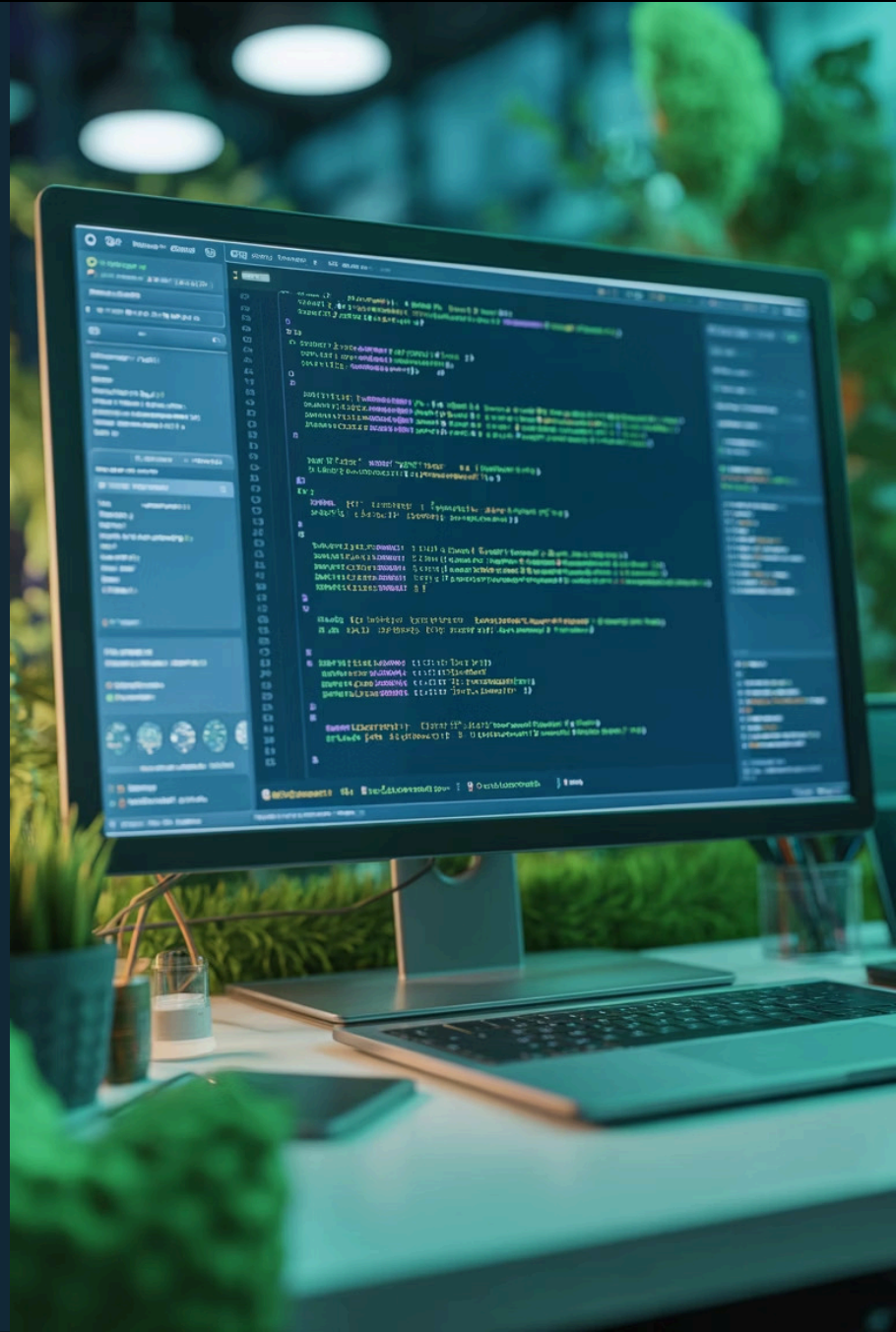
- CSV files with pd.read_csv()
- Excel files with pd.read_excel()
- SQL databases with pd.read_sql()
- Dask for large datasets

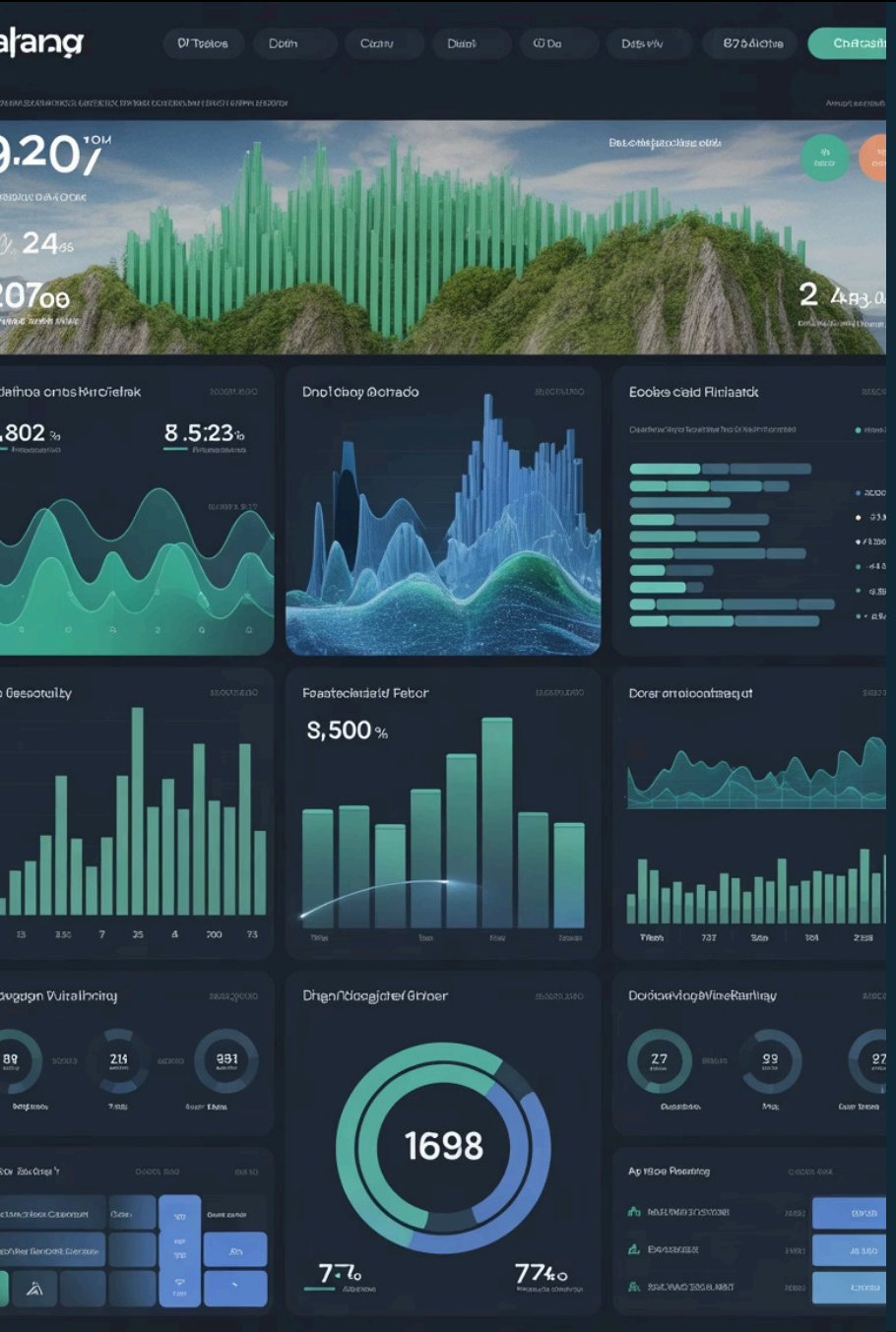
Data Type Conversions

- Numerical: pd.to_numeric(), astype(int)
- Categorical: astype('category')
- Datetime conversions
- Category code encoding

Missing Value Strategies

- Mean/median imputation for numerical
- Mode imputation for categorical
- Forward/backward fill for time series
- When NOT to impute (>50% missing, NMAR)





Visualization Techniques

Univariate Analysis

- Histograms for distribution
- Density plots (KDE)
- Count plots for categories
- Box plots for outliers

Bivariate Analysis

- Scatter plots for relationships
- Box plots across categories
- Violin plots for distributions
- Correlation heatmaps

Multivariate Analysis

- Pair plots for multiple variables
- Pivot table heatmaps
- Joint plots with distributions
- 3D visualizations

Advanced Techniques

- Row-wise percentages
- Column-wise percentages
- Grand total percentages
- Multiple aggregations

Expected Learning Outcomes

100%

Comprehensive EDA Skills

Master complete exploratory data analysis workflow on real sports datasets

8

Core Techniques

Essential steps from loading to advanced pivot table analysis

4

Python Libraries

Pandas, NumPy, Matplotlib, Seaborn proficiency

By completing this case study, you will develop a strong foundation in data analysis and visualization, equipping you with skills to analyze real-world datasets using Python. You'll learn to clean and transform data, identify and handle outliers effectively, use group by and pivot tables for complex aggregations, and visualize data to communicate insights clearly.

Start Your EDA Journey | Download Dataset